

ĐẠI HỌC BÁCH KHOA HÀ NỘI
TRƯỜNG CÔNG NGHỆ THÔNG TIN VÀ TRUYỀN THÔNG

— * —



BÁO CÁO BÀI TẬP

BTVN 26-05-2026

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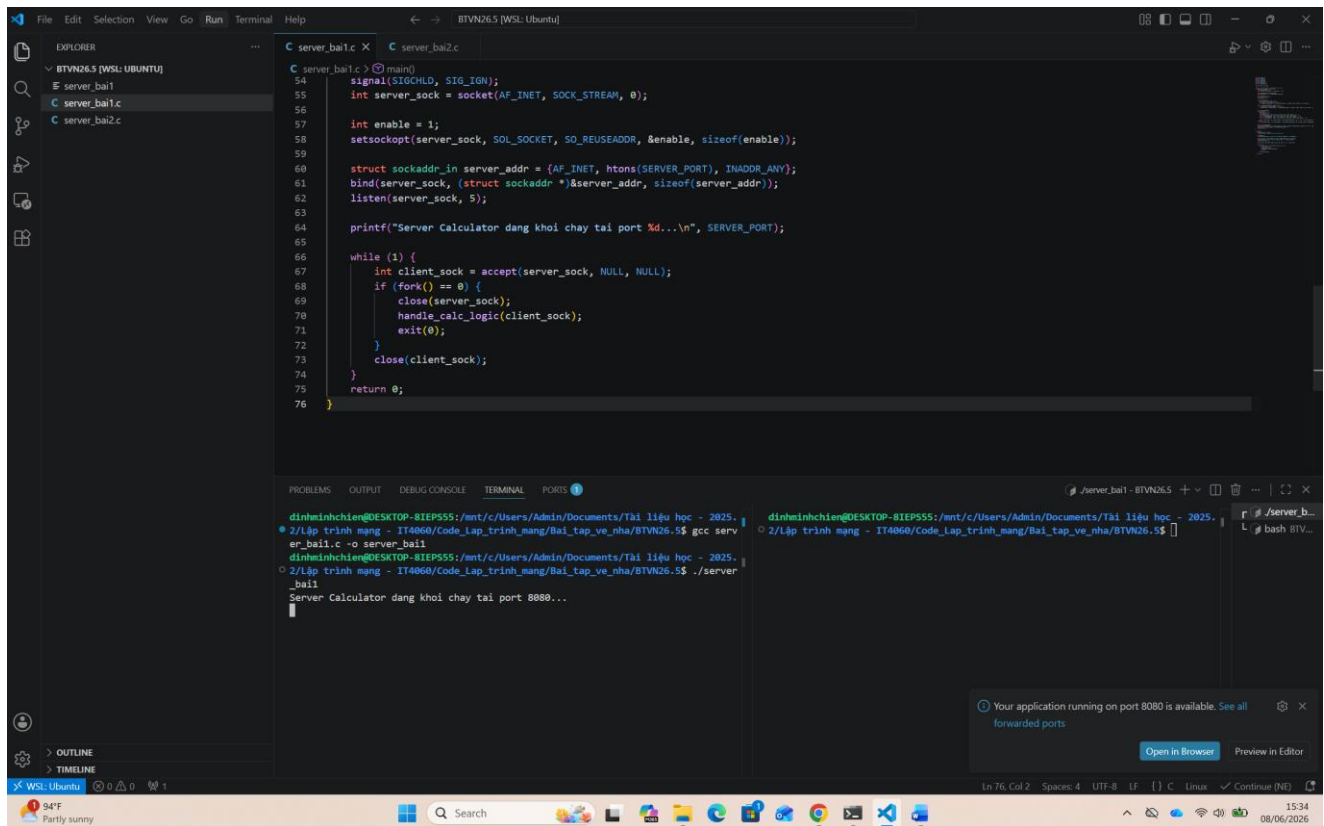
Mã Số Sinh Viên: 20235278

Học Phần: IT4060

Mã Lớp: 168503

B01 : HTTP_CALC_SERVER

Ảnh minh họa khởi tạo server:



The screenshot displays the Visual Studio Code (VS Code) interface with a C program named `server_bai1.c` open in the editor. The code is a simple HTTP server that listens on port 8080 and responds with "Server Calculator đang khởi chạy tại port 8080...". The code includes headers for `signal`, `unistd.h`, `sys/types.h`, `sys/socket.h`, `netinet/in.h`, and `stdio.h`. It uses `socket`, `bind`, `listen`, `accept`, and `fork` to handle incoming connections. The server is compiled and run in a terminal window within the VS Code environment. The terminal output shows the compilation command `gcc server_bai1.c -o server_bai1` and the execution command `./server_bai1`, which results in the message "Server Calculator đang khởi chạy tại port 8080...". A notification at the bottom right indicates that the application is running on port 8080 and is available for viewing in a browser.

```
server_bai1.c
1 // Khởi tạo server
2 #include <signal.h>
3 #include <unistd.h>
4 #include <sys/types.h>
5 #include <sys/socket.h>
6 #include <netinet/in.h>
7 #include <stdio.h>
8
9 int server_port = 8080;
10
11 int main()
12 {
13     signal(SIGCHLD, SIG_IGN);
14     int server_sock = socket(AF_INET, SOCK_STREAM, 0);
15
16     int enable = 1;
17     setsockopt(server_sock, SOL_SOCKET, SO_REUSEADDR, &enable, sizeof(enable));
18
19     struct sockaddr_in server_addr = {AF_INET, htons(server_port), INADDR_ANY};
20     bind(server_sock, (struct sockaddr *)&server_addr, sizeof(server_addr));
21     listen(server_sock, 5);
22
23     printf("Server Calculator đang khởi chạy tại port %d...\n", server_port);
24
25     while (1) {
26         int client_sock = accept(server_sock, NULL, NULL);
27         if (fork() == 0) {
28             close(server_sock);
29             handle_calc_logic(client_sock);
30             exit(0);
31         }
32         close(client_sock);
33     }
34     return 0;
35 }
```

```
./server_bai1
Server Calculator đang khởi chạy tại port 8080...
```

Ảnh minh họa client gửi lệnh GET:

```
C server_bai1.c X C server_bai2.c
C server_bai1.c > main()
54 signal(SIGCHLD, SIG_IGN);
55 int server_sock = socket(AF_INET, SOCK_STREAM, 0);
56
57 int enable = 1;
58 setsockopt(server_sock, SOL_SOCKET, SO_REUSEADDR, &enable, sizeof(enable));
59
60 struct sockaddr_in server_addr = {AF_INET, htons(SERVER_PORT), INADDR_ANY};
61 bind(server_sock, (struct sockaddr *)&server_addr, sizeof(server_addr));
62 listen(server_sock, 5);
63
64 printf("Server Calculator dang khoi chay tai port %d...\n", SERVER_PORT);
65
66 while (1) {
67     int client_sock = accept(server_sock, NULL, NULL);
68     if (fork() == 0) {
69         close(server_sock);
70         handle_calc_logic(client_sock);
71         exit(0);
72     }
73     close(client_sock);
74 }
75 return 0;
76 }
```

```
bash - BTYN26.5
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ gcc serv
er_bai1.c -o server_bai1
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ ./server
_bai1
Server Calculator dang khoi chay tai port 8080...
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ curl -v
"http://localhost:8080/?op=sub&a=20&b=8"
* Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 43716 failed: Connection refused
* Trying 127.0.0.1:8080...
* Connected to localhost (127.0.0.1) port 8080
> GET /?op=sub&a=20&b=8 HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
* no chunk, no close, no size. Assume close to signal end
<
* Closing connection
<html><body><h1>Ket qua: 20 - 8 = 12</h1></body></html>dinhhinhchien@DESKTOP-BIE
EP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$
```

Ảnh minh họa client gửi lệnh POST:

```
C server_bai1.c X C server_bai2.c
C server_bai1.c > main()
54 signal(SIGCHLD, SIG_IGN);
55 int server_sock = socket(AF_INET, SOCK_STREAM, 0);
56
57 int enable = 1;
58 setsockopt(server_sock, SOL_SOCKET, SO_REUSEADDR, &enable, sizeof(enable));
59
60 struct sockaddr_in server_addr = {AF_INET, htons(SERVER_PORT), INADDR_ANY};
61 bind(server_sock, (struct sockaddr *)&server_addr, sizeof(server_addr));
62 listen(server_sock, 5);
63
64 printf("Server Calculator dang khoi chay tai port %d...\n", SERVER_PORT);
65
66 while (1) {
67     int client_sock = accept(server_sock, NULL, NULL);
68     if (fork() == 0) {
69         close(server_sock);
70         handle_calc_logic(client_sock);
71         exit(0);
72     }
73     close(client_sock);
74 }
75 return 0;
76 }
```

```
bash - BTYN26.5
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ gcc serv
er_bai1.c -o server_bai1
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ ./server
_bai1
Server Calculator dang khoi chay tai port 8080...
dinhhinhchien@DESKTOP-BIEP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ curl -v
POST -d "op=add&a=10&b=5" http://localhost:8080/
* Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 43716 failed: Connection refused
* Trying 127.0.0.1:8080...
* Connected to localhost (127.0.0.1) port 8080
> POST -d "op=add&a=10&b=5" http://localhost:8080/ HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Content-Type: text/html
* no chunk, no close, no size. Assume close to signal end
<
* Closing connection
<html><body><h1>Ket qua: 10 + 5 = 15</h1></body></html>dinhhinhchien@DESKTOP-BIE
EP555:/mnt/c/Users/Admin/Documents/Tai lieu hoc - 2025.
2/Lap trinh mang - IT4060/Code_Lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$
```

Source code:

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
#include <signal.h>
#include <sys/socket.h>

#define SERVER_PORT 8080

void handle_calc_logic(int client_fd) {
    char raw_data[4096] = {0};
    recv(client_fd, raw_data, sizeof(raw_data) - 1, 0);

    char http_method[16], path_info[256];
    sscanf(raw_data, "%s %s", http_method, path_info);

    char operator[16] = "";
    double val1 = 0, val2 = 0;
    int is_valid = 0;

    if (strcmp(http_method, "GET") == 0) {
        char *q_ptr = strchr(path_info, '?');
        if (q_ptr && sscanf(q_ptr + 1, "op=%[^&]&a=%lf&b=%lf", operator, &val1, &val2) ==
3) is_valid = 1;
    }
    else if (strcmp(http_method, "POST") == 0) {
        char *body_ptr = strstr(raw_data, "\r\n\r\n");
        if (body_ptr && sscanf(body_ptr + 4, "op=%[^&]&a=%lf&b=%lf", operator, &val1,
&val2) == 3) is_valid = 1;
    }

    char html_output[1024];
    if (is_valid) {
        double calc_res = 0;
        char symbol = '?';
        if (strcmp(operator, "add") == 0) { calc_res = val1 + val2; symbol = '+'; }
        else if (strcmp(operator, "sub") == 0) { calc_res = val1 - val2; symbol = '-'; }
        else if (strcmp(operator, "mul") == 0) { calc_res = val1 * val2; symbol = '*'; }
        else if (strcmp(operator, "div") == 0 && val2 != 0) { calc_res = val1 / val2; symbol = '/'; }
    }

    snprintf(html_output, sizeof(html_output), "<html><body><h1>Ket qua: %.2g %c %.2g
= %.2g</h1></body></html>", val1, symbol, val2, calc_res);
    } else {
        snprintf(html_output, sizeof(html_output), "<html><body><h2>Thong bao: Sai cu phap.
Format: ?op=add&a=10&b=5</h2></body></html>");
    }
}

```

```

    char final_response[2048];
    snprintf(final_response, sizeof(final_response), "HTTP/1.1 200 OK\r\nContent-Type:
text/html\r\n\r\n%s", html_output);
    send(client_fd, final_response, strlen(final_response), 0);
    close(client_fd);
}

int main() {
    signal(SIGCHLD, SIG_IGN);
    int server_sock = socket(AF_INET, SOCK_STREAM, 0);

    int enable = 1;
    setsockopt(server_sock, SOL_SOCKET, SO_REUSEADDR, &enable, sizeof(enable));

    struct sockaddr_in server_addr = {AF_INET, htons(SERVER_PORT), INADDR_ANY};
    bind(server_sock, (struct sockaddr *)&server_addr, sizeof(server_addr));
    listen(server_sock, 5);

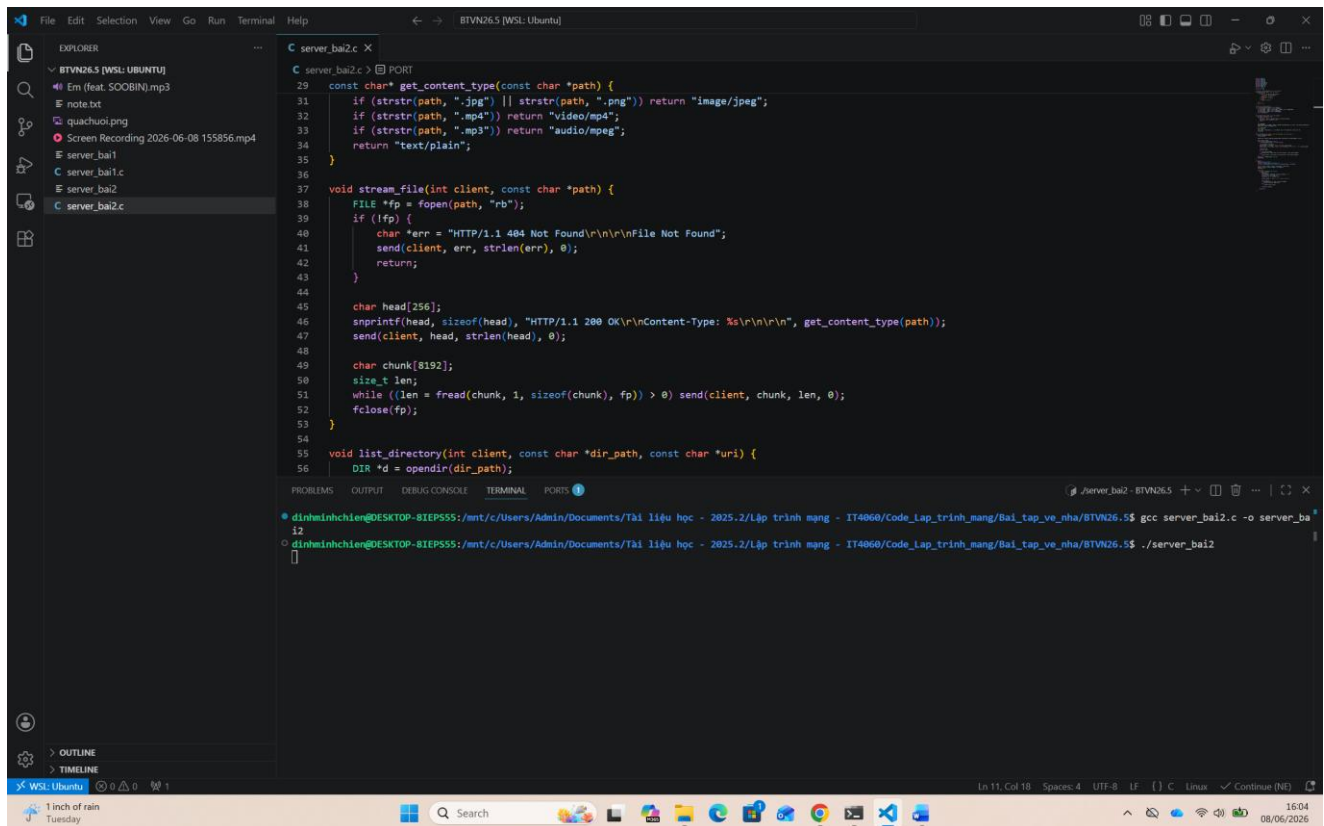
    printf("Server Calculator dang khoi chay tai port %d...\n", SERVER_PORT);

    while (1) {
        int client_sock = accept(server_sock, NULL, NULL);
        if (fork() == 0) {
            close(server_sock);
            handle_calc_logic(client_sock);
            exit(0);
        }
        close(client_sock);
    }
    return 0;
}

```

B02 : HTTP_FILE_SERVER

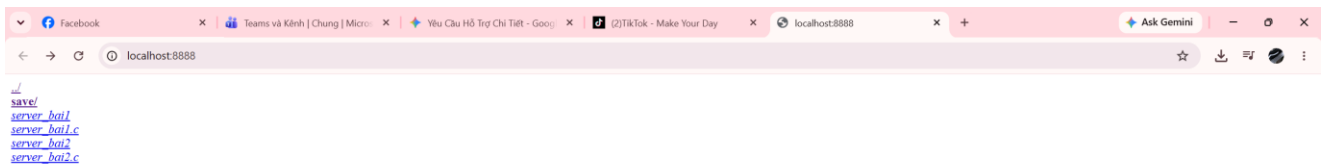
Ảnh minh họa khởi tạo server:



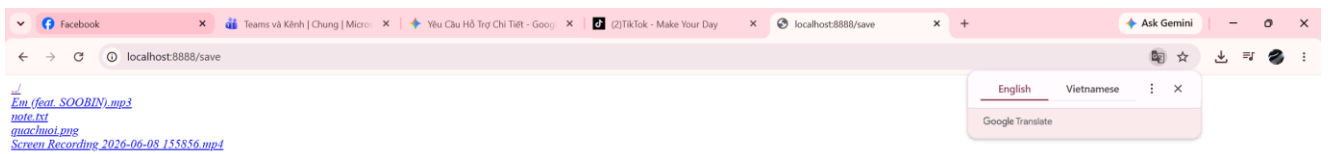
```
File Edit Selection View Go Run Terminal Help
BTVN26.5 [WSL: UBUNTU]
note.txt
quachuoi.png
Screen Recording 2026-06-08 155856.mp4
server_bai1
server_bai1.c
server_bai2
server_bai2.c

C server_bai2.c X
C server_bai2.c > PORT
29 const char* get_content_type(const char *path) {
31     if (strstr(path, ".jpg") || strstr(path, ".png")) return "image/jpeg";
32     if (strstr(path, ".mp4")) return "video/mp4";
33     if (strstr(path, ".mp3")) return "audio/mpeg";
34     return "text/plain";
35 }
36
37 void stream_file(int client, const char *path) {
38     FILE *fp = fopen(path, "rb");
39     if (!fp) {
40         char *err = "HTTP/1.1 404 Not Found\r\n\r\nFile Not Found";
41         send(client, err, strlen(err), 0);
42         return;
43     }
44
45     char head[256];
46     sprintf(head, "HTTP/1.1 200 OK\r\nContent-Type: %s\r\n\r\n", get_content_type(path));
47     send(client, head, strlen(head), 0);
48
49     char chunk[8192];
50     size_t len;
51     while ((len = fread(chunk, 1, sizeof(chunk), fp)) > 0) send(client, chunk, len, 0);
52     fclose(fp);
53 }
54
55 void list_directory(int client, const char *dir_path, const char *uri) {
56     DIR *d = opendir(dir_path);
57
58 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
dinhinhchien@DESKTOP-8TEPSS5:/mnt/c/Users/Admin/Documents/Tài liệu học - 2025.2/Lập trình mạng - IT4060/Code_lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ gcc server_bai2.c -o server_ba
i2
dinhinhchien@DESKTOP-8TEPSS5:/mnt/c/Users/Admin/Documents/Tài liệu học - 2025.2/Lập trình mạng - IT4060/Code_lap_trinh_mang/Bai_tap_ve_nha/BTVN26.5$ ./server_bai2
[]
Ln 11, Col 18 Spaces: 4 UTF-8 LF C Linux ✓ Continue (NE)
1 inch of rain Tuesday 16:04 08/06/2026
```

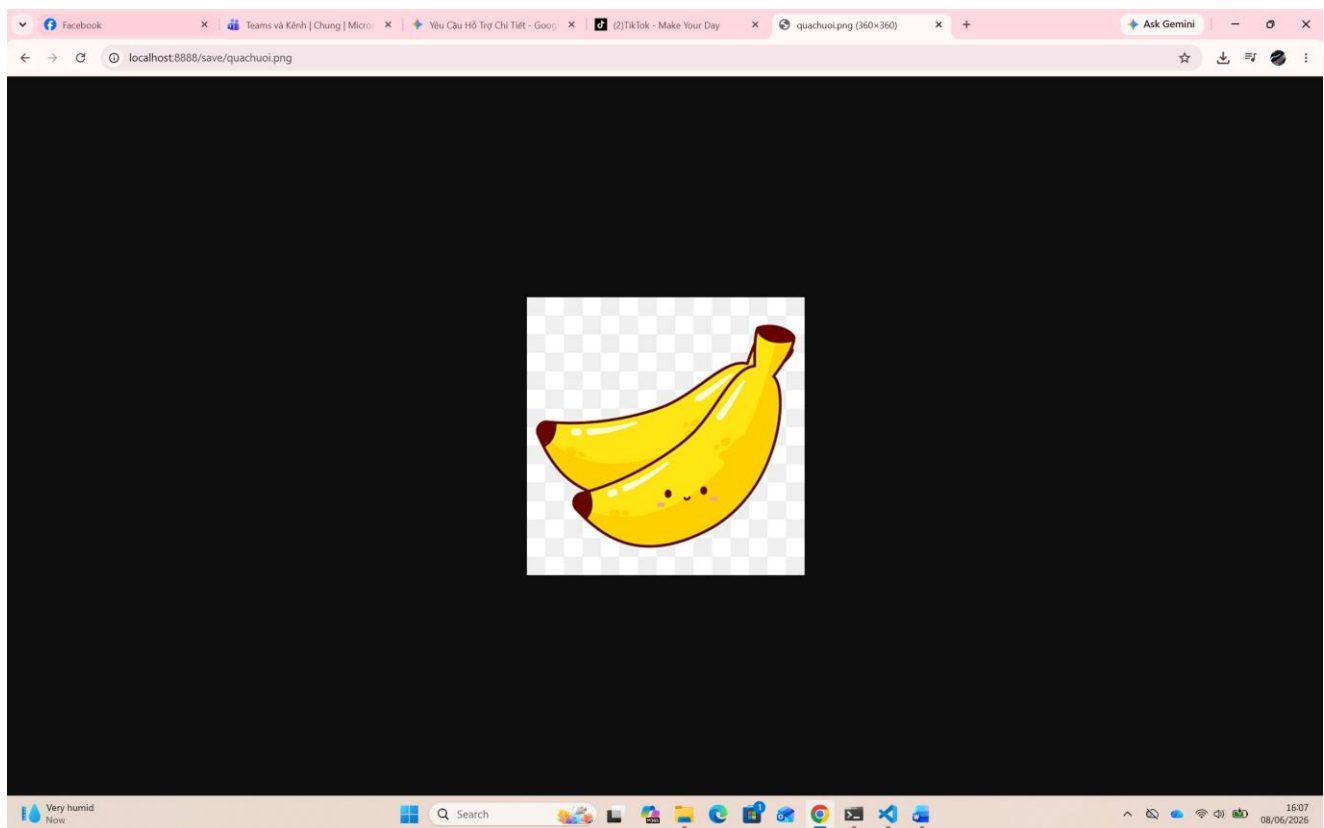
Ảnh minh họa giao diện web khi chạy server:



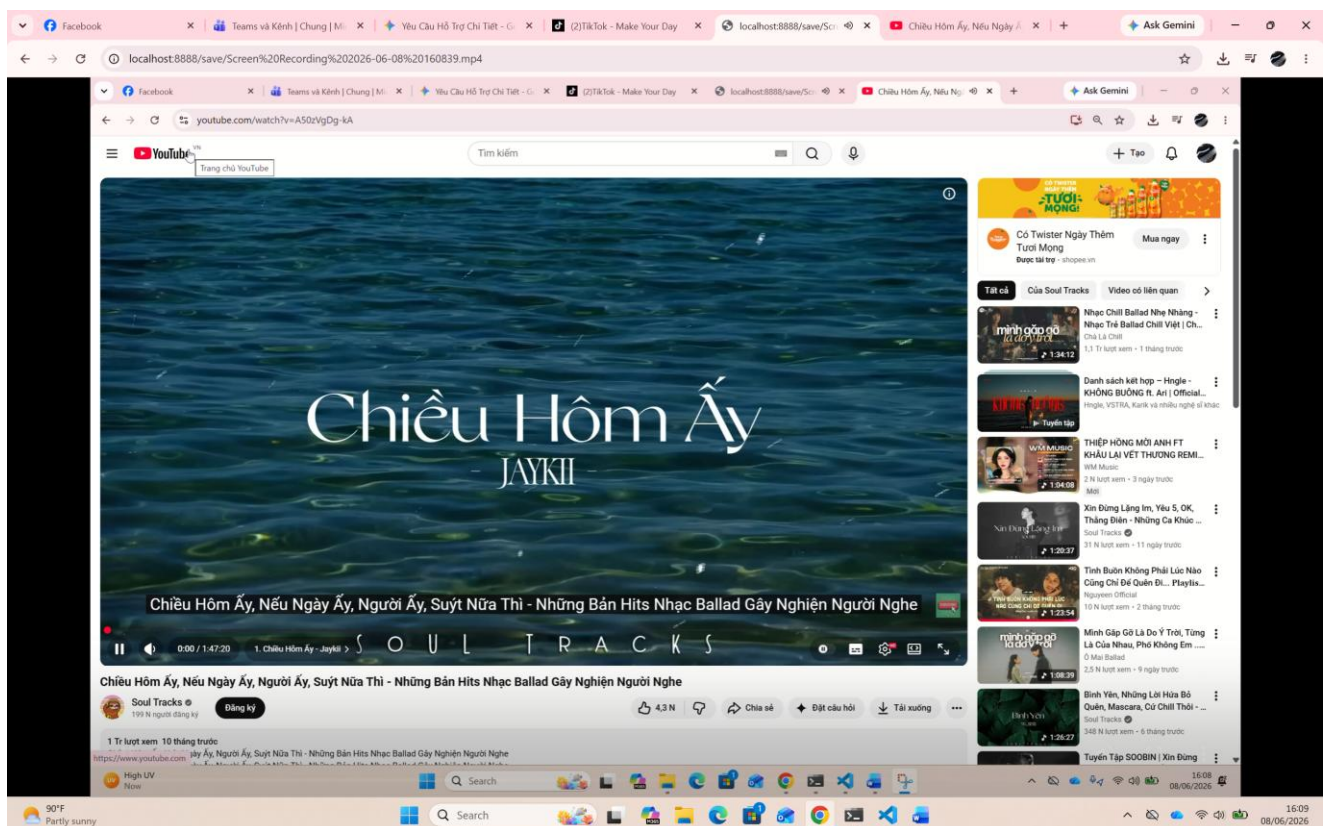
Ảnh minh họa khi ấn vào thư mục (hiển thị các file trong thư mục):



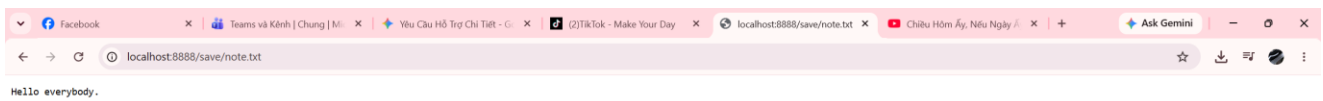
Ảnh minh họa khi ấn vào file ảnh:



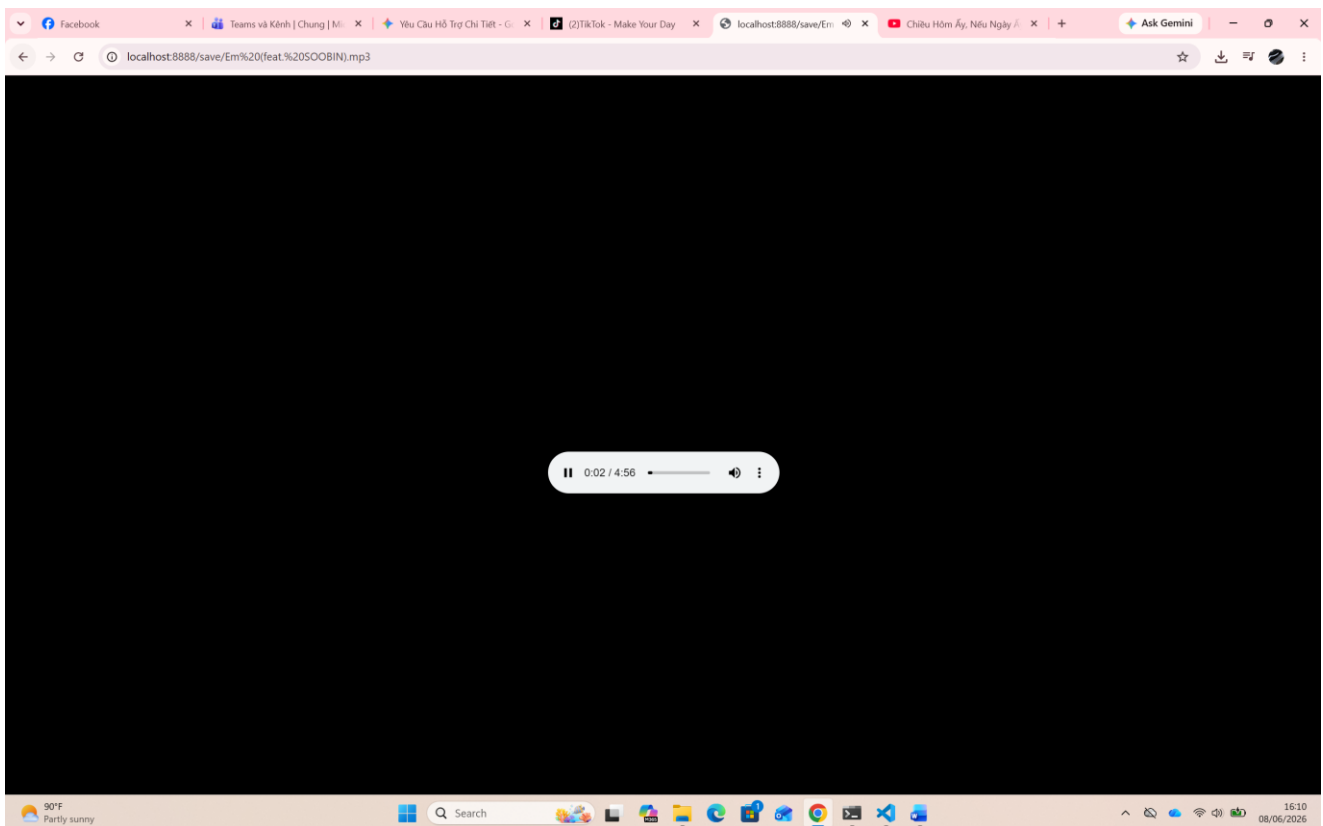
Ảnh minh họa khi ấn vào file video:



Ảnh minh họa khi ấn vào file văn bản:



Ảnh minh họa khi ấn vào file audio:



Source code:

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/socket.h>
#include <arpa/inet.h>
#include <sys/stat.h>
#include <dirent.h>
#include <signal.h>

#define PORT 8888

void clean_url(char *dest, const char *src) {
    while (*src) {
        if (*src == '%' && src[1] && src[2]) {
            *dest++ = ' '; src += 3;
        } else if (*src == '+') {
            *dest++ = ' '; src++;
        } else {
            *dest++ = *src++;
        }
    }
    *dest = '\0';
}

const char* get_content_type(const char *path) {
    if (strstr(path, ".html")) return "text/html";
    if (strstr(path, ".jpg") || strstr(path, ".png")) return "image/jpeg";
    if (strstr(path, ".mp4")) return "video/mp4";
    if (strstr(path, ".mp3")) return "audio/mpeg";
    return "text/plain";
}

void stream_file(int client, const char *path) {
    FILE *fp = fopen(path, "rb");
    if (!fp) {
        char *err = "HTTP/1.1 404 Not Found\r\n\r\nFile Not Found";
        send(client, err, strlen(err), 0);
        return;
    }

    char head[256];
    snprintf(head, sizeof(head), "HTTP/1.1 200 OK\r\nContent-Type: %s\r\n\r\n",
get_content_type(path));
    send(client, head, strlen(head), 0);

    char chunk[8192];
    size_t len;

```

```

while ((len = fread(chunk, 1, sizeof(chunk), fp)) > 0) send(client, chunk, len, 0);
fclose(fp);
}

void list_directory(int client, const char *dir_path, const char *uri) {
    DIR *d = opendir(dir_path);
    if (!d) return;

    send(client, "HTTP/1.1 200 OK\r\nContent-Type: text/html\r\n\r\n<html><body>", 55, 0);

    struct dirent *entry;
    while ((entry = readdir(d)) != NULL) {
        if (strcmp(entry->d_name, ".") == 0) continue;

        char full[1024], link[1024];
        snprintf(full, sizeof(full), "%s/%s", dir_path, entry->d_name);
        snprintf(link, sizeof(link), "%s%s%s", uri, (strcmp(uri, "/") == 0 ? "" : "/"), entry-
>d_name);

        struct stat st;
        stat(full, &st);

        if (S_ISDIR(st.st_mode))
            dprintf(client, "<b><a href='%s'>%s</a></b><br>", link, entry->d_name);
        else
            dprintf(client, "<i><a href='%s'>%s</a></i><br>", link, entry->d_name);
    }
    send(client, "</body></html>", 14, 0);
    closedir(d);
}

int main() {
    signal(SIGCHLD, SIG_IGN);
    int srv = socket(AF_INET, SOCK_STREAM, 0);
    int opt = 1; setsockopt(srv, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt));

    struct sockaddr_in addr = {AF_INET, htons(PORT), INADDR_ANY};
    bind(srv, (struct sockaddr *)&addr, sizeof(addr));
    listen(srv, 5);

    while (1) {
        int cli = accept(srv, NULL, NULL);
        if (fork() == 0) {
            close(srv);
            char req[2048], method[16], uri[256], path[512] = ".";
            recv(cli, req, sizeof(req), 0);
            sscanf(req, "%s %s", method, uri);

```

```
    clean_url(path + 1, uri);

    struct stat st;
    if (stat(path, &st) == 0 && S_ISDIR(st.st_mode))
        list_directory(cli, path, uri);
    else
        stream_file(cli, path);

    close(cli); exit(0);
}
close(cli);
}
}
```